

Request for Project Approval and Financial Support – SIH Disaster-Response Hexacopter

To,

1. Problem Statement

India is highly vulnerable to floods, cyclones, earthquakes, landslides and flash floods. These events can damage infrastructure, block roads, isolate communities and delay rescue operations. Ground-based assessment is often slow, risky and resource-intensive. During the first critical hours after a disaster, rescue teams require rapid information about survivor locations, fire, floodwater, debris, damaged structures, landslide zones, exposed electrical lines and possible hazardous-gas conditions. Limited or unavailable network connectivity further increases the difficulty of collecting and sharing field information.

2. Proposed Solution

We propose an autonomous aerodynamic hexacopter equipped with RGB, thermal, GPS/compass, IMU, optical-flow, 2D LiDAR, altitude-distance and gas-sensing modules. A Raspberry Pi 5 with a Raspberry Pi AI HAT+ 13 TOPS performs supported AI inference locally without depending on cloud connectivity. Pixhawk 4 manages flight stabilization, six-motor control, autonomous missions and safety functions. GPS, IMU, optical flow, RPLiDAR A1 and TFMMini support navigation, obstacle awareness, altitude assistance, local mapping and GPS-denied positioning experiments. The system detects survivors and visible hazards, geo-tags incidents, prioritizes emergency alerts, stores evidence locally and transmits results through local Wi-Fi or low-bandwidth RF to a laptop-based command dashboard.

3. Main System Modules

Flight and Control: Pixhawk 4, GPS/compass, RC transmitter and receiver – stabilization, six-motor control, waypoint missions, manual override and failsafe operation.

Onboard AI: Raspberry Pi 5 and Raspberry Pi AI HAT+ 13 TOPS – offline AI inference and sensor-data processing.

Multi-Sensor Detection: RGB camera, MLX90640 thermal sensor and MiCS-6814 gas sensor – people, heat signatures, visible hazards and gas-anomaly indication.

Navigation and Mapping: RPLiDAR A1, PMW3901 optical flow, TFMMini and IMU – obstacle scanning, 2D local mapping, altitude assistance and GPS-denied positioning support.

Communication and Dashboard: Local Wi-Fi/RF telemetry, laptop, Flask, JavaScript and map interface – alerts, mission monitoring, geo-tagged markers and situational reports.

Aerodynamic Hexacopter Design Note

The prototype uses a six-rotor aerodynamic configuration with a streamlined protective body, distributed motor arms, vibration isolation and shock-absorbing landing skids. The six-motor layout is intended to improve lift, stability and partial motor-failure tolerance compared with a basic quadcopter. The system is intended for controlled prototype demonstrations in challenging weather; it is not claimed to be certified for direct operation inside severe cyclone conditions.

4. Expected Outcomes

- Rapid aerial assessment of disaster-affected locations.
- On-device survivor and hazard detection without Internet dependency.
- GPS-based geo-tagging of detected incidents and evidence.
- 2D LiDAR-assisted local mapping and obstacle awareness.
- Prioritized alerts for survivors, fire, floodwater, debris, damaged structures and gas anomalies.
- Local command-center dashboard with mission status, map markers and situational reports.
- A functional prototype suitable for controlled demonstrations and further research.

5. Estimated Budget with Purpose

No.	Component	Qty.	Amount (₹)	Purpose
1	Custom carbon-fiber/composite hexacopter frame aerodynamic body	1	7,000	Aerodynamic body and six-arm structure; stability and redundancy
2	Pixhawk 4 flight controller	1	33,840	Flight stabilization, six-motor control, autonomous mission and failsafe
3	Raspberry Pi 5 4GB (already available)	1	0	Onboard AI, mapping, sensor fusion and communication
4	Raspberry Pi AI HAT+ 13 TOPS	1	7,469	AI acceleration for offline inference
5	RGB camera	1	3,500	Visual detection of people and visible hazards
6	MLX90640 thermal sensor	1	5,000	Heat-signature sensing and human-presence confirmation
7	RPLiDAR A1M8 2D LiDAR	1	8,630	360° obstacle scanning, local mapping and 2D SLAM support
8	TFMini / TFMMini-S distance sensor	1	3,500	Downward or forward distance, altitude and landing assistance
9	PMW3901 optical-flow sensor	1	3,260	Local position estimation during GPS-denied operation
10	BN-880 GPS + compass	1	2,060	Position, heading and geo-tagging
11	MiCS-6814 gas sensor module	1	800	Gas-anomaly sensing and possible chemical-release indication
12	BLDC motors	6	3,900	Six-rotor propulsion for lift, stability and partial motor-failure tolerance
13	ESCs	6	4,500	Independent electronic speed control for all six motors
14	Matched propellers	1 set	1,200	Six matched propellers plus spare set
15	4S Li-Po battery	1	5,500	Main power source
16	Li-Po balance charger	1	2,000	Safe battery charging
17	Power module/PDB/BEC/regulators	1 set	3,000	Power distribution and voltage regulation
18	RC transmitter and receiver	1 set	4,000	Manual control, testing and emergency override
19	Local Wi-Fi/RF telemetry pair	1 pair	2,151	Local alerts and low-bandwidth telemetry
20	MicroSD/onboard storage	1	1,500	AI models, images, maps, logs and reports
21	Pixhawk buzzer, LEDs, wiring and connectors	1 set	1,579	Status indication and electrical connections
22	Sensor mounts and protective enclosure	1 set	1,500	Secure and protected sensor installation
23	Spare connectors, vibration isolation and safety items	1 set	2,000	Reliability, maintenance and safety
24	Testing, transport and contingency	1	5,000	Controlled flight testing, calibration and unforeseen expenses

TOTAL ESTIMATED PROJECT COST: ₹1,12,889

Requested institutional support: ₹1,13,000

Budget Note: The above amounts are estimated planning rates and may vary according to vendor quotations, availability, GST and final specifications. Procurement will be carried out only after institutional approval and according to college purchase procedures. The six-motor design requires six motors, six ESCs and six matched propellers, which are included in this updated budget. The TFMini item may be replaced by an available upgraded equivalent after compatibility verification.

6. Request

We kindly request the college to review and approve the attached project concept and sanction financial support of ₹1,13,000 for procurement, assembly, software integration, calibration, testing and validation of the SIH disaster-response hexacopter prototype.

We will maintain proper records of procurement and expenditure and use the approved resources exclusively for the project and competition requirements.

Thank you for considering our proposal. We look forward to your support and approval.

Yours faithfully,

Name: _____

Programme / Department: _____

Team / Project: SIH Autonomous Edge-AI Disaster-Response Hexacopter

Contact: _____

Faculty Mentor: _____

7. Technical Coverage and Design Clarification

- Six-rotor aerodynamic hexacopter: six BLDC motors, six ESCs and six matched propellers are included.
- On-device AI: Raspberry Pi 5 with AI HAT+ 13 TOPS processes supported optimized models locally.
- Navigation: Pixhawk 4, GPS/compass, IMU, PMW3901 optical flow, RPLiDAR A1 and TFMini support autonomous navigation, obstacle awareness, altitude assistance and GPS-denied positioning experiments.
- Hazard sensing: RGB, MLX90640 and MiCS-6814 provide visual detection, thermal confirmation and gas-anomaly indication.
- Communication: local Wi-Fi/Rf telemetry sends alerts and mission information to the laptop dashboard; the drone stores data when communication is unavailable.
- Operational limitation: the aerodynamic body improves stability and wind handling, but cyclone-level operation requires separate wind-tunnel, structural and flight-safety validation.